Adjusted-R^2

As a penalty for adding regressors to increase the R^2 value, the concept of adjusted-R^2, denoted by $\bar{R}^2$ has been introduced.

$$\bar{R}^2 = 1 - \frac{Residual\ sum\ of\ squares/n-k}{total\ \ sum\ of\ squares/n-1} = 1 - (1 - \bar{R}^2)\frac{n-1}{n-k} \quad \dots (8.21)$$

where *n* is the sample size, and *k* is the number of explanatory variables. From (8.21) we see that $\bar{R}^2 \leq R^2$. It indicates that the adjusted-R^2 penalizes for adding regressors. There is an increase in the value of Adjusted-R^2 only if the absolute *t*-value of the added regressor is greater than 1. Hence $\bar{R}^2$ ensures meaningful and valid comparisons. Remember that comparison between models is possible for the same dependent variables. If dependent variables are different, we cannot compare among models.

8.5.2 The AIC and SIC Criteria

The Akaike Information Criterion (AIC) criterion is based on the maximum likelihood estimation of regression models. We consider a small number of regression models based on economic theory. We find out the value of the log-likelihood function of each model. The AIC criterion is based on the idea of imposing a penalty for adding regressors.

$$AIC = e^{2k/n}\frac{\sum \hat{u}_i^2}{n} = e^{2k/n}\frac{residual\ sum\ of\ squares}{n} \quad \dots (8.22)$$

where *k* is the number of regressors (including the intercept) and *n* is the number of observations.

For mathematical convenience, we can write

$$ln\ AIC = \left(\frac{2k}{n}\right) + ln\left(\frac{residual\ sum\ of\ squares}{n}\right) \quad \dots (8.23)$$

where $ln\ AIC =$ natural log of AIC and $\left(\frac{2k}{n}\right) =$ penalty factor.

We find out the value of '*ln AIC*' according to equation (8.23) for all the contending models.We compare the value of '*ln AIC'* of the models, and the model with the lowest value of AIC is preferred. The advantages of AIC over the coefficient of determination (R^2) are as follows: (i) AIC criterion is useful for not only the in-sample but also the out-of-sample forecasting performance. (ii) AIC criterion can be used for both nested and non-nested models. (iii) AIC criterion can also help in determining the lag length in an AR(p) model.

Schwartz Information Criterion (SIC)

The SIC criterion is also based on the method of maximum likelihood estimation. It is also called the Bayesian Information Criterion (BIC), or the Schwartz Bayesian Criterion (SBC). The SIC criterion, as in the case of adjusted-R^2 and the AIC criterion, penalises regression models having larger number of regressors.

The SIC is defined as:

$$SIC = n^{k/n}\frac{\sum \hat{u}^2}{n} = n^{k/n}\frac{residual\ sum\ of\ squares}{n} \quad \dots (8.24)$$

The above can be expressed in log-form as

$$\ln \text{SIC} = \frac{k}{n}\ln n + \ln\frac{\textit{residual sum of squares}}{n} \qquad \dots (8.25)$$

where $[(k/n)\ln n]$ is the penalty factor. A lower value of SIC implies a better model. The SIC, like AIC, can also be used for comparison of both in-sample and out-of-sample forecasting performance of the model.

8.5.3 Mallow's C_p Criterion

Mallow's C_p criterion for model selection is defined as

$$C_p = \frac{RSS_p}{\hat{\sigma}^2} - (n - 2p) \qquad \dots (8.26)$$

where *n* is the number of observations, total regressors are *k* out of which only p are included in the model and RSS_p is the residual sum of squares obtained from p (included) regressors. $E(\hat{\sigma}^2)$ is the unbiased estimator of the true σ^2. If the model with only *p* regressors is the true model, then $E(RSS_p) = (n-p)\sigma^2$. Therefore, (8.26) is approximately equal to

$$E(C_p) \approx \frac{(n-p)\sigma^2}{\sigma^2} - (n - 2p) \approx p \qquad \dots (8.27)$$

While choosing a regression model according to the C_p criterion, a model with a lower C_p value is selected, about equal to *p*. To put it differently, following the principle of parsimony, we will choose a model with *p* regressors (p<k) that gives a fairly good fit to the data.

8.6 CAUTION ABOUT MODEL SELECTION CRITERIA

We highlighted earlier that econometric models should be grounded in economic theory and reasoning. Hence, when refining an econometric model, it is important to consider the theoretical justification for including or omitting a variable. To ensure a properly defined model, it is essential to go through earlier studies to grasp the theoretical principles. Additionally, the quality of the model we develop is contingent upon the quality of the data we gather. If the data collected is free from issues like multicollinearity or autocorrelation, we are more likely to end up with a stronger model.

Many a time, we observe certain relationship between two variables. Such relationships, however, may be superficial or spurious. Let us take an example. At a traffic light, cars stop when the signal is red. It does not mean that cars cannot move when there is a red light in front of them. It also does not mean that traffic lights have some damaging effect on moving cars. The reason is observance of traffic rules. Unless we look into the traffic rules and go by observation only, our reasoning will be wrong. The dependent variable and the independent variable both may be affected by another variable. In such cases the relationship is 'confounded'. Confounded relationship indicates a situation where a variable, Z, affects two other variables, X and Y, in such a way that we observe certain association between X and Y. In the example given above traffic rules affect the working of traffic lights as well as car drivers.

You should note one more issue regarding selection of econometric models. Different test criteria may suggest different models. For example, economic logic suggests that there could be two possible econometric models (say, model A and model B) for a particular issue. You may come across a situation such that $\bar{R}^2$ test suggests model A and AIC criterion suggest model B. In such situations you should carry out several tests and then only chose the best model.

Adjusted-R^2, Mallows C_p, p-values, etc. may point to different regression equations without much clarity to the econometrician. Thus, we conclude that none of the methods for model selection listed above are adequate by itself. There is no substitute to theoretical understanding of the related literature, accurately collected data, practical understanding of the problem, and common sense while specifying an econometric model.

CHECK YOUR PROGRESS 2

1) Explain how the RESET is applied.

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2) What are the limitations of RESET?

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3) What precaution one should one take while dealing with model selection criteria?

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8.7 LET US SUM UP

Correct specification of an econometric model determines the accuracy of the estimates obtained. Therefore, correct specification of an econometric model is very important. Economic theory and logic guide us in specification of econometric models. In order to correctly specify an econometric model, all relevant explanatory variables should be included in the model. No relevant explanatory variable should be excluded from the model. Further, the functional form of the model should be correct.

Selection of an appropriate econometric model is a difficult task. We have to take into account economic theory and logic behind the econometric model. There could be many competing models for a particular issue. There are certain criteria on the basis of which the best econometric model is selected. These criteria could be $\bar{R}^2$, AIC, SIC, and Mallow's C_p.We have described the formulae for these test criteria in the Unit.

8.8 KEY WORDS

Bias : The difference between the expected value of an estimator and the population parameter that the estimator is supposed to be estimating.

RESET : RESET is used to test for the presence of error in model specification, particularly its functional form. The RESET is based on F-test.

AIC Criterion : If there are k number of explanatory variables in a regression model, then $ln\, AIC = \left(\frac{2k}{n}\right) + ln\left(\frac{residual\ sum\ of\ squares}{n}\right)$. The AIC criterion is used in model selection among competing regression models.

SIC Criterion : The SIC criterion is used in model selection among competing regression models. If n is the sample size and k is the number of regressors in a model, then SIC = $n^{k/n}\frac{\sum \hat{u}^2}{n} = n^{k/n}\frac{residual\ sum\ of\ squares}{n}$

8.9 SOME USEFUL BOOKS

Dougherty, C. (2011). *Introduction to Econometrics*, Fourth Edition, Oxford University Press

Gujarati, D N and D C Porter (2009), *Basic Econometrics*, Fifth Edition, McGraw Hill

Johnston, J., Dinardo, J. (2007), "*Econometric Methods*", 4th Edition, McGraw-Hill Education (Asia), International Edition, pp. 109-125

Maddala, G. S. and K. Lahiri (2009), *Introduction to Econometrics*, Fourth Edition, John Wiley and Sons

Wooldrige, Jeffrey (2022), *Introductory Econometrics*, Seventh Edition, Cengage Learning

8.10 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) The basic assumptions of the classical regression model are as follows:

   a) The regression model is linear in parameters

   a) $E(X_i u_i) = 0$ (regressor is non-stochastic)

   b) $E(u_i) = 0$

   c) $E(u_i)^2 = \sigma^2$

   d) $E(u_i u_j) = 0$ for $i \neq j$

   e) The explanatory variables (X_i) are independent of one another.

2) Go through Section 8.2. It is important because incorrect specification has serious implications on desirable properties of the estimators. Therefore, inferences drawn from mis-specified models are not correct.

3) The major sources of specification error are: (i) inclusion of redundant/ superfluous/ irrelevant variables, (ii) omission of relevant variables, (iii) incorrect functional form, (iv) incorrect assumption regarding the error term, and (v) error in measurement of variables.

4) Due to inclusion of irrelevant variables in a model, we obtain unbiased but inefficient least squares estimators of the parameters. The larger variance reduces the precision of the estimates resulting in wider confidence intervals. This may lead to type II error.

5) When a relevant variable is dropped from the model, we obtain biased estimators of the parameters and there is a decline if the efficiency of the estimators. The estimator of error variance is higher, as a result of which tests of hypothesis are invalid. Therefore, the inferences drawn are faulty.

Check Your Progress 2

1) Go through Section 8.4 and answer.

2) The RESET leaves us with much ambiguity. It does not tell us the next course of action.

3) Go through Section 8.6 and answer.

UNIT 9 AUTOCORRELATION*

Structure

9.0 OBJECTIVES

After going through this unit, you should be in a position to

- identify the problem of autocorrelation in regression analysis,
- explain how the problem of autocorrelation affects the estimators of the regression model parameters,
- explain the various methods available to detect autocorrelationin a dataset, and
- suggest remedial measures for the problem of autocorrelation.

9.1 INTRODUCTION

An importantassumption of the classical regression model is that the error terms u_i are independent of one another. However, this may not be true always and in certain cases the error terms may be related to one another. In notations,

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$$E(u_i, u_j) \neq 0; i \neq j \qquad \ldots(9.1)$$

We find several examples in our day to day life where the successive observations appear to follow a natural order over time. The upward or downward trend in time series data generally has an in-built momentum which makes successive observations interdependent. Hence, the successive error terms may be correlated. For example, the BSE Sensex may be seen moving upwards or downwards for successive days. In this Unit we deal mainly with correlation between error terms in time series data.

9.2 CONCEPT OF AUTOCORRELATION

There are two ways in which the error terms may be correlated:

Spatial correlation: It is the correlation between error terms in cross-sectional data. For instance, recall the demonstration effect, which says that consumption behaviour of households is affected by that of their neighbours (at times termed as "keeping up with the Joneses" effect). Households try to match or keep up to the standards of their neighbours. Thus, in a household study, the error terms may be found to be correlated. In the present Unit, however, we do not discuss issues related to spatial correlation.

Serial correlation or Autocorrelation: In time series data, the correlation between error terms is known as autocorrelation or serial correlation. There are various reasons why time series data may exhibit autocorrelation. The series may be *non-stationary* (mean, variance and covariance do not remain constant over time) which causes the error terms to be correlated. Omission of a relevant variable from the model is also captured by the error term causing them to be correlated. Even specifying an incorrect functional form (for instance, fitting a linear cost function instead of a cubic function) will cause the error terms to be correlated with one another. Sometimes the explained variable depends on its lagged values. For instance, current consumption depends on past consumption as consumption habits are slow to change. In such cases we need to include lagged value of consumption as explanatory variable. If the lagged variable is not included, the error term may capture that effect resulting in correlation between the error terms. Interpolation and extrapolation of data may also cause residuals to follow a systematic pattern.

Technically, $u_{t-2}, u_{t-1}, u_t, u_{t+1}, u_{t+2}$ .... may be correlated to one another. Here, 'u' denotes the error term and the subscript denotes the time period. We have changed the subscripts to 't' keeping the view the fact that autocorrelation takes place mostly in time series data. First order autocorrelation is correlation between u_t and u_{t-1}. Second order autocorrelation is correlation between u_t and u_{t-2} and so on. Autocorrelation of order 'k' is the correlation between $u_t$and u_{t-k}. For 'n' observations, there could be $(n-1)$ autocorrelations.

9.3 CONSEQUENCES OF AUTOCORRELATION

In the presence of autocorrelation, the OLS estimators are:

- Linear
- Unbiased
- Asymptotically normally distributed

The variance of the estimators, however, gets affected. Thus in the presence of autocorrelation, the OLS estimator $\hat{\beta}$ is not the best linear unbiased estimator (BLUE). It implies that the usual tests of significance discussed in Units 3 and 4 cannot be applied. We discuss further on each of the above propositions.

9.3.1 Unbiasedness

Let $y_t = \beta x_t + u_t$ … (9.2)

Let us denote the OLS estimator of β by $\hat{\beta}$.

We know from Unit 3 that

$$\hat{\beta} = \frac{\sum x_t y_t}{\sum x_t^{\,2}} = \beta + \frac{\sum x_t u_t}{\sum x_t^{\,2}} \quad \text{… (9.3)}$$

In the presence of autocorrelation, the assumptions $E(u_t) = 0$ and $E(x_t u_t) = 0$ continue to be true.

Thus, $E\left(\frac{\sum x_t u_t}{\sum x_t^{\,2}}\right) = 0$

Hence, $E(\hat{\beta}) = \beta$

That is, $\hat{\beta}$ is unbiased.

9.3.2 Minimum Variance

In order to derive the variance of $\hat{\beta}$ in the presence of autocorrelation, let us consider a two-variable regression model $y_t = \beta x_t + u_t$ and assume that the error terms are auto-correlated. In notations,

$E(u_t, u_{t+s}) \neq 0 \ (s \neq 0)$

We assume that the error term follows a '**first order auto-regressive**' or **AR(1) process** such that

$u_t = \rho u_{t-1} + \varepsilon_t$ …(9.4)

In the above AR(1) process, we assume that current error (u_t) equals certain proportion of previous error (u_{t-1}) plus a random shock (ε_t). Here, 'ρ' is the *coefficient of autocovariance or first order coefficient of autocorrelation.* The random shock ε_t satisfies the usual OLS assumptions.

$E(\varepsilon_t) = 0$

$Var(\varepsilon_t) = \sigma_\varepsilon^{\,2}$, and

$Cov(\varepsilon_t, \varepsilon_{t+s}) = 0;\ s \neq 0$ …(9.5)

The random shock (ε_t) which satisfies the above properties is usually called the **white noise**. The error terms u_t can be correlated in many forms. For instance, an AR(2) process is given by

$u_t = \rho_1 u_{t-1} + \rho_2 u_{t-1} + \varepsilon_t$

Here u_t depends upon error terms of previous two periods. Similarly, an AR(3) process is given by

$u_t = \rho_1 u_{t-1} + \rho_2 u_{t-1} + \rho_3 u_{t-2} + \varepsilon_t$

and so on.

Sometimes, the error term u_t follows a Moving Average (MA) process. For instance, a first order moving average process is given by MA(1) where,

$u_t = \varepsilon_t + \rho_1 \varepsilon_{t-1}$ … (9.6)

Here u_t depends upon the white noise of previous period. We will discuss further on AR and MA processes in Unit 18 of this course. In order to keep matters simple, in the present Unit, we assume that the error term follows an AR(1) process.

(a) $\boldsymbol{Var(u_t)}$

For an AR(1) process the variance of the error term is given by,

$var(u_t) = var\ (\rho u_{t-1} + \varepsilon_t) = \rho^2 var\ (u_{t-1}) + var(\varepsilon_t)$

Recall from equation (9.3) that $var(\varepsilon_t) = {\sigma_\varepsilon}^2$.

Since $var\ (u_t) = var\ (u_{t-1}) = \sigma^2$, the above equation can be written as

$\sigma^2 = \rho^2\sigma^2 + {\sigma_\varepsilon}^2$

On re-arranging terms in the above equation we obtain

$\sigma^2 - \rho^2\sigma^2 = {\sigma_\varepsilon}^2$

or, $\sigma^2(1-\rho^2) = {\sigma_\varepsilon}^2$

This gives us, $var(u_t) = \sigma^2 = \frac{{\sigma_\varepsilon}^2}{(1-\rho^2)}$ …(9.7)

In equation (9.7), the variance of the error term (u_t) is homoscedastic even though the error terms are correlated.

(b) $\boldsymbol{Cov\ (u_t, u_{t-1})}$

Covariance betweenu_t and u_{t-1} is given by

$Cov\ (u_t, u_{t-1}) = E\{u_t - E(u_t)\}\{u_{t-1} - E(u_{t-1})\}$

Since $E(u_t) = E(u_{t-1}) = 0$, the covariance of the error term, $Cov\ (u_t, u_{t-1})$, is given by

$Cov\ (u_t, u_{t-1}) = E(u_t u_{t-1})$

On successive substitution we find that

$$E(u_t u_{t-1}) = E(\rho u_{t-1}{}^2 + \varepsilon_t u_{t-1}) = \rho E(u_{t-1}{}^2) = \rho \frac{\sigma_\varepsilon{}^2}{(1-\rho^2)} \qquad \text{... (9.8)}$$

[Notice that in an AR(1) process u_{t-1} is not influenced by ε_t as the former occurs in the earlier time period;thus $E(\varepsilon_t u_{t-1}) = 0$]

Similarly, $cov\ (u_t, u_{t-2}) = \rho^2 \frac{\sigma_\varepsilon{}^2}{(1-\rho^2)}$

In general terms we can write that

$$cov\ (u_t, u_{t-s}) = \rho^s \frac{\sigma_\varepsilon{}^2}{(1-\rho^2)} \qquad \text{… (9.9)}$$

We need to assume here that |ρ| <1, otherwise the variance and covariance given above are undefined. In case |ρ| ≥1, we encounter is the popular 'unit root problem' (see Unit 17 for further discussion on this).

(c) $\boldsymbol{Var(\hat{\beta})_{AR(1)}}$

We know from Unit 3 that in the classical regression model the OLS estimator for variance of $\hat{\beta}$, say $Var(\hat{\beta})_{OLS} = \frac{\sigma^2}{\sum x_t{}^2}$

The underlying assumption is that the error terms are not correlated, i.e.,

$E(u_t, \varepsilon_{t+s}) = 0; s \neq 0$

When this assumption is violated, and u_t follows an AR process, the variance and covariance of u_t are as follows:

$$var(u_t) = \frac{\sigma_\varepsilon{}^2}{(1-\rho^2)}$$

$$cov\ (u_t, u_{t-s}) = \rho^s \frac{\sigma_\varepsilon{}^2}{(1-\rho^2)}$$

Recall that the OLS estimator of β, that is, $\hat{\beta} = \frac{\sum x_t y_t}{\sum x_t{}^2} = \beta + \frac{\sum x_t u_t}{\sum x_t{}^2}$

Hence, $var(\hat{\beta} - \beta) = var\left(\frac{\sum x_t u_t}{\sum x_t{}^2}\right) = E(\frac{\sum x_t u_t}{\sum x_t{}^2})^2$

For AR(1) process, $var(\hat{\beta})_{AR(1)}$is given by

$$E(\frac{\sum x_t u_t}{\sum x_t{}^2})^2 = \frac{1}{(\sum x_t{}^2)^2} var(\sum x_t\ u_t) \qquad \text{... (9.10)}$$

We retain the assumption that variable x_t is non-stochastic and $E(x_t u_t) = 0$.

Thus, $E(\frac{\sum x_t u_t}{\sum x_t{}^2})^2 =$

$$\frac{1}{(\sum x_t{}^2)^2} (\sigma^2 \sum x_t{}^2 + 2Cov(u_t, u_{t-1}) \sum x_t\ x_{t-1} + 2Cov(u_t, u_{t-2}) \sum x_t\ x_{t-2} + ...)$$